

LESSON 6-4 • SESSION 1 • PRACTICE • FORM B

Writing Numerical Expressions

Learning target – I can write a numerical expression that matches a situation. • 6.EE.A.2

REMEMBER

- A **numerical expression** has numbers and operations but **no equal sign**.
- “Each” or “per” means **equal groups**, so multiply. Totals of groups are **added**. “5 less than 12” is $12 - 5$.
- Use **parentheses** for a part done first: the sum of 4 and 6, tripled, is $3 \times (4 + 6)$.

WORKED EXAMPLE

Ms. Ortiz buys 4 pizzas at \$9 each and one \$6 salad. Write an expression for the total. Evaluate it.

Step 1 Pizzas are equal groups: 4×9 .

Step 2 Add the salad: $4 \times 9 + 6$.

Step 3 Evaluate: $36 + 6 = 42$.

Answer $4 \times 9 + 6 = \$42$

WORDS **numerical expression** (*expresión numérica*) • **evaluate** (*evaluar*) • **product** (*producto*)

PRACTICE

show your work • label units

- Write a numerical expression for each phrase. Do not evaluate.
 - 6 more than the product of 7 and 2 _____
 - 18 less than 60 _____
 - the sum of 3 and 7, doubled _____
 - the quotient of 56 and 7, minus 3 _____
- Art Club Order.** An art club buys 6 sets of brushes at \$11 a set and 3 packs of canvas at \$8 a pack. It also lays out mosaic tiles in 9 rows of 9.
 - Write one expression for the total supply cost. Evaluate it. _____ dollars
 - Write the mosaic tiles as a power. Evaluate it. _____ tiles

Show your work.

- Field Trip.** A field trip costs \$12 for each of 25 students, plus one \$30 bus fee. Write one expression for the total cost. Evaluate it.
Hint One product for the students, then add the fee.
- Square Garden.** A square garden has 14 plants along each side, filling the whole square. Write the number of plants as a power. Evaluate it.

Show your work.

Answer _____ dollars

Hint A square uses the same number twice.

Show your work.

Answer _____ plants

- Pretzel Bags.** Mia fills 4 bags with 15 pretzels each, then eats 6 pretzels. She wrote $4 + 15 - 6$ for the pretzels left. Justify or correct her expression.

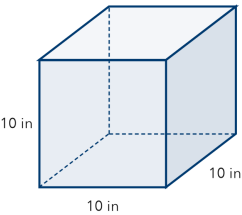
Hint Which operation matches the word “each”?

Answer _____ pretzels

6. **Cube Bin.** A cube-shaped bin is 10 inches on every edge. Write the number of 1-inch cubes that fill it as a power. Evaluate it.

Hint A cube has 3 equal dimensions.

Show your work.



Answer _____ cubes

7. **Supply Change.** A class buys 10 notebooks at \$4 each and 15 pens at \$3 each. It pays with \$100.
- Write an expression for the total cost. _____
 - Write an expression for the change. Evaluate it. _____ dollars
 - Why do parentheses help in part b? _____

8. **Tickets and Programs.** A group buys 5 tickets at \$8 each and 4 programs at \$5 each. Write one expression for the total cost. Evaluate it.

Hint Write a product for each item, then add.

Show your work.

Answer _____ dollars

FIND THE ERROR

explain the mistake, then fix it

9. **Jaden's Work.** Jaden counts seedlings in 6 trays of 8 seedlings each, plus 9 extra seedlings. He wrote $6 + 8 + 9 = 23$. Explain his mistake. Then write and evaluate a correct expression.

TEST PRACTICE

choose the best answer

10. "The sum of 8 and 2, doubled" is written –
- A** $8 + 2 \times 2$ **B** $2 \times (8 + 2)$ **C** $8 \times 2 + 2$ **D** $2 + 8 \times 2$
11. Parentheses tell you which part to –
- A** do first **B** skip **C** do last **D** always multiply

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

60 S	1,000 E	196 M	13 Q	54 B	28 X	330 U	30 Z	90 N	15 R
#2a		#3	#4	#5	#6	#7b	#8		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____